are commonly seen in tropical rain forests. Examples: Orchids, Lianas, Hanging Mosses and Money plant.

Morphological adaptations

- Root system is extensively developed. These roots may be of two types. They are Clinging roots and Aerial roots.

  Clinging roots fix the epiphytes firmly on the surface of the supporting objects.

  Aerial roots are green coloured roots which may hang downwardly and absorb moisture from the atmosphere with the help of a spongy tissue called **velamen**.
- Stem of some epiphytes are succulent and develop pseudobulb or tuber.
- Generally the leaves are lesser in number and may be fleshy and leathery
- **Myrmecophily** is a common occurrence in the epiphytic vegetation to prevent the predators.
- The fruits and seeds are very small and usually dispersed by wind, insects and birds.

Anatomical adaptations

- Multilayered epidermis is present. Inner to the velamen tissue, the peculiar exodermis layer is present.
- Presence of thick cuticle and sunken stomata greatly reduces transpiration.
- Succulent epiphytes contain well developed parenchymatous cells to store water.

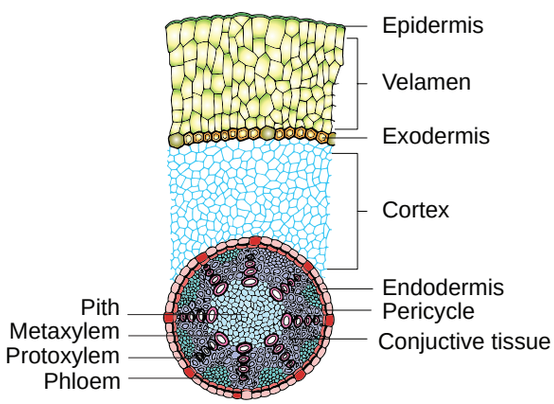


Figure 6.25: T.S. of an aerial root of *Orchid* showing velamen tissue

Physiological adaptations

Special absorption processes of water by velamen tissue .

Halophytes

There are special type of **Halophytic plants** which grow on soils with high concentration of salts. Examples: *Rhizophora, Sonneratia* and *Avicennia*.

Halophytes are usually found near the sea-shores and Estuaries. The soils are physically wet but physiologically dry. As plants cannot use salt water directly they require filtration of salt using physiological processes. This vegetation is also known as **mangrove forest** and the plants are called **mangroves.**

Morphological adaptations

- The temperate halophytes are herbaceous but the tropical halophytes are mostly bushy
- In addition to the normal roots, many stilt roots are developed
- A special type of negatively geotropic roots called **pneumatophores** with **pneumathodes** to get sufficient aeration are also present. They are called breathing roots. Example: *Avicennia*

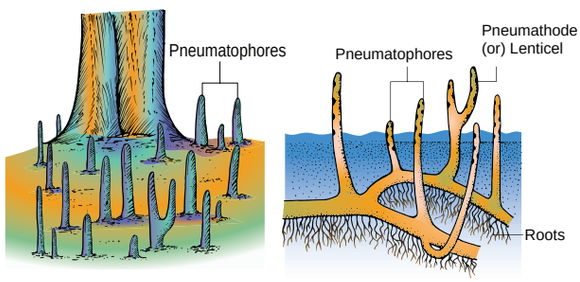


Figure 6.26a: Pneumatophores of mangrove plant

- Presence of thick cuticle on the aerial parts of the plant body
- Leaves are thick, entire, succulent and glossy. Some species are **aphyllous** (without leaves).

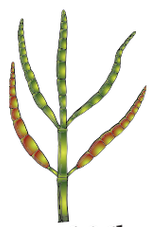


Figure 6.26b: Succulent halophyte - *Salicornia*

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